

Modeling with Systems and Matrices

Answer Key

Algebra 2

Quick Answers

- | | |
|---------------------------------------|------------------------------------|
| 1. $a = 75, s = 45$ | 7. $x = 3, y = 5, -10$ |
| 2. $x = 8, y = 4$ | 8. $x = 3, y = 4, x = -4, y = -3$ |
| 3. $x = 2, y = 5, 7$ | 9. $x = 1500, y = 2500$ |
| 4. $x = 12, y = 24, -11$ | 10. 0, |
| 5. $x = 2, y = 3, x = 4, y = 7, 2, 4$ | 11. $x = 7, y = 10, x = 10, y = 7$ |
| 6. $c = 1, k = 5$ | 12. $x = 80, y = 80, z = 40$ |
-

Curriculum

Level: Algebra 2

HSA-REI.C.6 – *problems 1, 2, 6, 9, 12*

HSA-REI.C.7 – *problems 5, 8, 11*

HSN-VM.C.6–C.12 – *problems 3, 4, 7, 10*

Difficulty 2–5 of 5 across 18 tagged checks.

Common wrong answers

4. *If they got 11: swapped the diagonals and computed bc minus ad*
7. *If they got 10: swapped the diagonals and computed bc minus ad*
-

1. School play: adult tickets \$8, student tickets \$5, 120 tickets sold, \$825 collected. Find how many of each were sold.

Solution: Let a = adult tickets, s = student tickets.

$$a + s = 120, \quad 8a + 5s = 825$$

Multiply the count equation by 5 and subtract it from the money equation:

$$(8a + 5s) - (5a + 5s) = 825 - 600 \implies 3a = 225 \implies a = 75$$

Back-substitute: $s = 120 - 75 = 45$. Check: $8(75) + 5(45) = 600 + 225 = 825$ ✓

$a = 75, s = 45$

2. Acid mixture: a 20% solution and a 50% solution combine to make 12 L of a 30% solution. Find the litres of each.

Solution: Let x = litres of the 20% solution, y = litres of the 50% solution. Volume gives $x + y = 12$; acid content (in percent-litres) gives $20x + 50y = 30(12) = 360$.

$$\begin{aligned} 20x + 20y &= 240 && \text{(volume equation, times 20)} \\ (20x + 50y) - (20x + 20y) &= 360 - 240 \implies 30y = 120 \end{aligned}$$

So $y = 4$ and $x = 12 - 4 = 8$. Check: $20(8) + 50(4) = 160 + 200 = 360$ ✓

$x = 8 \text{ L}, y = 4 \text{ L}$

3. Print shop: 3 pens + 2 markers cost \$16; 4 pens + 5 markers cost \$33. (a) Write $AX = B$ and find $\det A$. (b) Solve for the two prices.

Solution (a):

$$A = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 16 \\ 33 \end{bmatrix}, \quad \det A = (3)(5) - (2)(4) = 15 - 8$$

$\det A = 7$

Solution (b): Since $\det A \neq 0$ the inverse exists: $A^{-1} = \frac{1}{7} \begin{bmatrix} 5 & -2 \\ -4 & 3 \end{bmatrix}$, so

$$X = A^{-1}B = \frac{1}{7} \begin{bmatrix} 5(16) - 2(33) \\ -4(16) + 3(33) \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 14 \\ 35 \end{bmatrix}$$

Check: $3(2) + 2(5) = 16$ and $4(2) + 5(5) = 33$ ✓ (pens \$2 a box, markers \$5 a box).

$x = 2, y = 5$

4. Tour company: 2 vans + 3 buses seat 96; 5 vans + 2 buses seat 108. (a) Find $\det A$. (b) Solve by Cramer's rule.

Solution (a): The coefficient matrix is $A = \begin{bmatrix} 2 & 3 \\ 5 & 2 \end{bmatrix}$, so $\det A = (2)(2) - (3)(5) = 4 - 15$.

$\det A = -11$

Solution (b): Replace a column of A with B and divide by $\det A$:

$$\begin{aligned} x &= \frac{\begin{vmatrix} 96 & 3 \\ 108 & 2 \end{vmatrix}}{\det A} = \frac{192 - 324}{-11} = \frac{-132}{-11} = 12 \\ y &= \frac{\begin{vmatrix} 2 & 96 \\ 5 & 108 \end{vmatrix}}{\det A} = \frac{216 - 480}{-11} = \frac{-264}{-11} = 24 \end{aligned}$$

Check: $2(12) + 3(24) = 24 + 72 = 96$ ✓ and $5(12) + 2(24) = 60 + 48 = 108$ ✓

$x = 12, y = 24$

5. Drone path $y = x^2 - 4x + 7$ meets cable $y = 2x - 1$. Find the meeting points.

Solution: Both expressions equal y , so set them equal — this is the substitution step that turns a nonlinear system into one quadratic equation:

$$\begin{aligned}x^2 - 4x + 7 &= 2x - 1 \\x^2 - 6x + 8 &= 0 \\(x - 2)(x - 4) &= 0\end{aligned}$$

So $x = 2$ or $x = 4$. Back-substitute into the line: $y = 2(2) - 1 = 3$ and $y = 2(4) - 1 = 7$.

$(2, 3) \quad \text{and} \quad (4, 7)$

6. Kayak: 12 km downstream in 2 hours, 12 km upstream in 3 hours. Find the kayak speed k and current speed c in km/h.

Solution: Downstream the speeds add, upstream they subtract, and distance = rate $\times$ time:

$$2(k + c) = 12 \implies k + c = 6, \quad 3(k - c) = 12 \implies k - c = 4$$

Adding the two reduced equations gives $2k = 10$, so $k = 5$, and then $c = 6 - 5 = 1$. Check: downstream 6 km/h for 2 h is 12 km $\checkmark$; upstream 4 km/h for 3 h is 12 km $\checkmark$

$k = 5, \quad c = 1$

7. Workshop: chairs need 4 cutting and 6 finishing hours, tables need 3 cutting and 2 finishing hours; 27 cutting and 28 finishing hours are used. (a) Find $\det A$. (b) Solve for the numbers of chairs and tables.

Solution (a): $A = \begin{bmatrix} 4 & 3 \\ 6 & 2 \end{bmatrix}$, so $\det A = (4)(2) - (3)(6) = 8 - 18$.

$\det A = -10$

Solution (b): By Cramer's rule,

$$\begin{aligned}x &= \frac{\begin{vmatrix} 27 & 3 \\ 28 & 2 \end{vmatrix}}{\det A} = \frac{54 - 84}{-10} = \frac{-30}{-10} = 3 \\y &= \frac{\begin{vmatrix} 4 & 27 \\ 6 & 28 \end{vmatrix}}{\det A} = \frac{112 - 162}{-10} = \frac{-50}{-10} = 5\end{aligned}$$

Check: $4(3) + 3(5) = 27 \checkmark$ and $6(3) + 2(5) = 28 \checkmark$

$x = 3, \quad y = 5$

8. Flight path $y = x + 1$ crosses the boundary $x^2 + y^2 = 25$. Find the crossing points.

Solution: Substitute the line into the circle:

$$\begin{aligned}x^2 + (x + 1)^2 &= 25 \\2x^2 + 2x + 1 &= 25 \\x^2 + x - 12 &= 0 \\(x + 4)(x - 3) &= 0\end{aligned}$$

So $x = 3$ or $x = -4$, and $y = x + 1$ gives $y = 4$ or $y = -3$. Check: $3^2 + 4^2 = 25$ ✓ and $(-4)^2 + (-3)^2 = 25$ ✓

$$\boxed{(3, 4) \quad \text{and} \quad (-4, -3)}$$

9. \$4000 split between a 5% account and an 8% account earns \$275 in one year. Find the two amounts.

Solution: Let x be the amount at 5% and y the amount at 8%. Multiplying the interest equation by 100 keeps the arithmetic in whole numbers:

$$x + y = 4000, \quad 5x + 8y = 27500$$

Substituting $x = 4000 - y$ into the second equation:

$$5(4000 - y) + 8y = 27500 \implies 20000 + 3y = 27500 \implies 3y = 7500$$

So $y = 2500$ and $x = 4000 - 2500 = 1500$. Check: $0.05(1500) + 0.08(2500) = 75 + 200 = 275$ ✓

$$\boxed{x = 1500, \quad y = 2500}$$

10. Caterer: 2 sandwich trays + 3 fruit trays = 12 servings; 4 sandwich trays + 6 fruit trays = 30 servings. (a) Find $\det A$. (b) Solve or show there is no solution. (c) Explain what the determinant says.

Solution (a): $A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$, so $\det A = (2)(6) - (3)(4) = 12 - 12$.

$$\boxed{\det A = 0}$$

Solution (b): A zero determinant means A has no inverse, so the matrix method stops here and we compare the equations directly. Doubling the first equation gives $4x + 6y = 24$, but sheet two claims $4x + 6y = 30$. The same combination of trays cannot serve both 24 and 30 people, so the system is inconsistent: the two lines are parallel and never meet.

$$\boxed{\text{no solution}}$$

Solution (c) — model answer (open response, review by hand): $\det A = 0$ says the two rows of A are multiples of each other: sheet two orders exactly twice everything on sheet one. So the two sheets describe the *same* recipe scaled up, and their serving totals must scale the same way. Twice 12 is 24, not 30, so the two sheets contradict each other and at least one was written down wrong. (Had sheet two read 24 servings, the determinant would still be 0, but then the two sheets would agree and there would be infinitely many solutions instead of none.)

11. Garden: 34 m of edging encloses 70 m². Find the side lengths.

Solution: Let x and y be the side lengths.

$$2x + 2y = 34 \implies x + y = 17, \quad xy = 70$$

Substitute $y = 17 - x$ into the area equation:

$$x(17 - x) = 70 \implies x^2 - 17x + 70 = 0 \implies (x - 7)(x - 10) = 0$$

So $x = 7$ (then $y = 10$) or $x = 10$ (then $y = 7$) — the same rectangle either way. Check: perimeter $2(7) + 2(10) = 34$ ✓, area $7 \times 10 = 70$ ✓

$$\boxed{7 \text{ m} \times 10 \text{ m}}$$

12. Concert: adult \$12, student \$8, child \$5; 200 tickets sold for \$1800, with 2 times as many student tickets as child tickets. Find all three counts.

Solution: Let x, y, z be the adult, student and child ticket counts.

$$x + y + z = 200, \quad 12x + 8y + 5z = 1800, \quad y = 2z$$

Use the third equation to eliminate y everywhere:

$$\begin{aligned} x + 2z + z &= 200 \implies x = 200 - 3z \\ 12(200 - 3z) + 8(2z) + 5z &= 1800 \\ 2400 - 36z + 16z + 5z &= 1800 \\ -15z &= -600 \end{aligned}$$

So $z = 40$, then $y = 2(40) = 80$ and $x = 200 - 3(40) = 80$. Check: $80 + 80 + 40 = 200$ ✓ and $12(80) + 8(80) + 5(40) = 960 + 640 + 200 = 1800$ ✓

$$\boxed{x = 80, y = 80, z = 40}$$